HTML :

🧩 **STEP 1: Create HTML Web Page**

Open Visual Studio Code

Create file:

index.html

Paste:

```
<!DOCTYPE html>
<html>
<head>
  <title>Conference Info</title>
</head>
<body>

<h1>Tech Conference 2026</h1>

<h2>Schedule</h2>
<ul>
  <li>10:00 AM - Opening</li>
  <li>11:00 AM - AI Session</li>
</ul>

<h2>Speakers</h2>
<ul>
  <li>John Doe</li>
  <li>Jane Smith</li>
</ul>

<h2>Venue</h2>
<p>Chennai Trade Center</p>
```

```
<h2>Contact</h2>
<p>Email: conf@gmail.com</p>

</body>
</html>
```

🌐 STEP 2: Create GitHub Repository

Go to GitHub

- Create repo: conference-project

💻 STEP 3: Clone Repository

```
git clone https://github.com/dhanush270605/conference-project.git
cd conference-project
code .
```

📩 STEP 4: Add HTML File

👉 Place index.html inside project folder

🔁 STEP 5: Commit and Push

```
git add .
git commit -m "added html page"
git push origin main
```

🐳 STEP 6: Install Docker

Install Docker

Check:

```
docker --version
```

📦 STEP 7: Create Dockerfile

Dockerfile

```
FROM nginx:alpine
COPY . /usr/share/nginx/html/
CMD ["nginx", "-g", "daemon off;"]
```

🔨 **STEP 8: Build Docker Image**

```
docker build -t conference-app .
```

▶️ **STEP 9: Run Container**

```
docker run -d -p 8081:80 conference-app
```

Open:

http://localhost:8081

☁️ **STEP 10: Push Docker Image**

Go to Docker Hub

```
docker login
docker tag conference-app dhanush270605/conference-app
docker push dhanush270605/conference-app
```

⚙️ **STEP 11: Install Jenkins**

Install Jenkins

Open:

http://localhost:8080

🔁 **STEP 12: Create Jenkins Pipeline**

Pipeline Name:

conference-pipeline

Script:

```
pipeline {
    agent any

    stages {
```

```
        stage('Clone Code') {
            steps {
                git branch: 'main', url: 'https://github.com/dhanush270605/conference-project.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t conference-app .'
            }
        }

        stage('Run Container') {
            steps {
                bat 'docker run -d -p 8082:80 conference-app'
            }
        }
    }
}
```

Click **Build Now**

☸️ STEP 13: Enable Kubernetes

Open Docker Desktop
👉 Settings → Kubernetes → Enable

📄 STEP 14: Create Deployment File

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
```

```
  name: conference-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: conference
  template:
    metadata:
      labels:
        app: conference
    spec:
      containers:
      - name: conference
        image: dhanush270605/conference-app
        ports:
        - containerPort: 80
```

📄 **STEP 15: Create Service File**

service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: conference-service
spec:
  type: NodePort
  selector:
    app: conference
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

🚀 **STEP 16: Deploy to Kubernetes**

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

🔍 **STEP 17: Check Status**

kubectl get pods

kubectl get services

🌐 **STEP 18: Open Application**

Option 1:

http://localhost:30007

Option 2 (Working method you used):

kubectl port-forward deployment/conference-deployment 8085:80

Open:

http://localhost:8085

📊 **STEP 19: Visualize**

Open Docker Desktop → Kubernetes tab
See:

- Pods ✅
- Services ✅
- Deployment ✅

⚠️ **COMMON ERRORS (REAL ONES YOU FACED)**

- ❌ your-username used → fixed
- ❌ master vs main → fixed
- ❌ Docker not running → fixed
- ❌ ImagePullBackOff → fixed
- ❌ Port conflict (8080 Jenkins) → fixed

🎯 **FINAL FLOW**

HTML → GitHub → Docker → Docker Hub → Jenkins → Kubernetes → Browser

---

🎉 **FINAL STATUS**

👉 Your project is:

- ✅ Fully working
- ✅ Deployed
- ✅ Production-like setup
- ✅ Ready for viva

🚀 **If you want next**

I can give:

- ✅ Viva questions + answers
- ✅ Diagram for presentation
- ✅ Add CI/CD auto deploy (Jenkins → Kubernetes)

Just tell 👍